



A 4-inch flexible OLED display of QVGA resolution with a curvature radius of 2 inches was showcased at CES



Ennova announced a USB flash drive that packs a OLED screen which also doubles up as a fingerprint scanner

How LCD panels work

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Liquid Crystal Displays (LCDs) have their advantages for both TVs and computer monitors. However, there are two things they haven't done as well as CRT; high contrast and colour depth/accuracy.

This however is set to change with the introduction of LED back-lights. To see why requires a look at how 'traditional' LCDs work.

LCD panels use the variable polarisation of electronically controlled liquid crystal cells to vary the amount of polarised light that will pass through the panel. In other words, an LCD panel is composed of a large number (a typical 1,600 by 1,200 computer monitor has 5,760,000 cells) of tiny, voltage controlled, light valves. The cells themselves are neutral density; they pass or block all frequencies of visible light more or less equally. To provide the sensation of variable colour each pixel consists of three of these cells, with individual red, green and blue filters placed in front of each group of three cells. A wide range of colours can be produced by varying the relative amounts

RGB LED backlights deliver brighter, darker and more colourful displays

of light passing through each of the three cells making up a pixel. Ambient light can be reflected from a plate behind the cells, or a back-light can be used for a brighter, emissive, display. Most displays used these days have a back-light and until recently this would have been a Cold Cathode Fluorescent Lamp (CCFL)

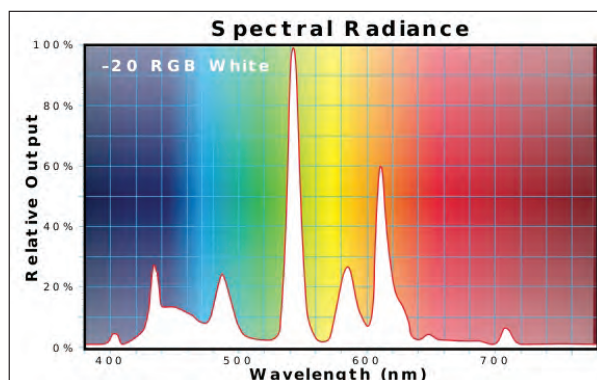
One of the weaknesses of LCDs is low contrast. When the liquid crystal cells are fully open (in the off state with no voltage applied), they do not transmit all of the light from the back-light, for one thing the light passes through a polarising filter, so

at least half of the original light from the back-light is blocked. However, more significantly, when the cells are closed they do not block all of the light. The ratio of light passed, between the on and off state defines the displays contrast ratio (note – contrast is always a ratio, not an absolute measurement). In the older type of LCD panel the CCFL back-light remains on at a fixed brightness all the time (it may turn off during power saving) and has no effect on the contrast ratio, which is determined entirely by the on/off ratio of the liquid crystal panel (ignoring the effect of ambient light reflected from the internal structure of the LCD for the moment).

The Fluorescent lamp

Fluorescent lamps for general lighting were introduced commercially in 1938 and were first used in large quantities to light factories during the Second World War (commercial LEDs became available in 1968 and blue LEDs were not introduced until the late 1990's). Fluorescent lamps are significantly more efficient than incandescent lamps and small fluorescent lamps (Compact Fluorescent Lamps with electronic ballasts – CFL) have been heavily marketed in recent years as 'energy saving' lamps. Conversion efficiency of electrical power to radiant power varies between 17 and 21 per cent compared to less than 5 per cent for the filament incandescent lamp. The luminous efficiency of CFL sources is typically 60 to 72 lumens per watt, compared to 8 to 17 lumens per watt for incandescent lamps.

In back-light applications, efficiency (regardless of the type of light source) is low because a lot of light is lost passing through the diffusers, polarisers and colour filters, perhaps only 5 per cent of the light from the



Spectral plot of a CCFL back-light taken from the manufacturers brochure. The light output is extremely uneven with large spikes at various frequencies

back-light reaches the viewer.

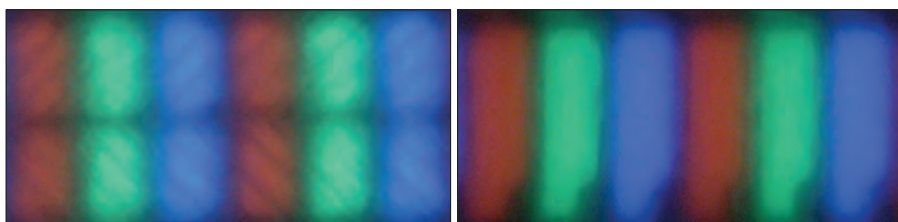
Very recently there has been something of a backlash against fluorescent lamps since they contain mercury, which is poisonous and tends to get released into the environment when old lamps are thrown away.

Fluorescent lamps work by ionising a column of low pressure gas (mercury vapour) causing it to glow with ultra-violet light. The inside of the glass tube containing the gas is coated with a mixture of phosphors that react to the ultra-violet light by emitting light in the visible spectrum. Various mixtures of phosphors are used in the tubes to give an approximation of 'white' light with some variation. For example "Warm White" tubes have more red while "Cool White" tubes have more blue. The spectral output of a fluorescent lamp is surprisingly uneven.

The CCFLs used for LCD back-lights use a high voltage to start the ionisation of the gas while the tubes used for general lighting run at lower voltages and use a momentary heating coil. As far as the colour range or gamut of LCDs is concerned, it is the uneven spectral response and the range of light frequencies emitted by the tube phosphors that limit performance. Early CCFLs used similar phosphors to the

Jargon Buster

- Polarization - a property of waves that describes the orientation of their oscillations.
- Ionization - is the physical process of converting an atom or molecule into an ion by adding or removing charged particles such as electrons or other ions.
- Dead pixels - They could occur on LCDs when the screen is damaged or put under pressure.



Photomicrographs of two pixels from two LCD panels from different manufacturers. The images are blurred because they are shot through the anti-glare front diffuser. Nevertheless it is still possible to see that the two panels use different cell geometries